2-THE ENDODERM : INTRODUCTION & GENERAL INFORMATION 1 ## INTRODUCTION

DURATION : 11:08

STUDY SHEET

COURSE SUMMARY

This video offers an in-depth introduction to the endoderm, an essential embryonic tissue, in relation to the mesoderm which can protect it. Key concepts include the organisation and flexion of the embryo, the importance of respiration, and the role of the digestive system in our overall health. The teaching is oriented towards osteopathy and integrates practical elements, such as the link between the digestive tract and emotions, waste elimination, as well as the impact of gut flora on our well-being. By exploring the link with seasons and fasting, the video highlights the need to take care of our body and mind to achieve optimal balance.

THE ENDODERM: INTRODUCTION AND GENERALITIES

We will study an **embryonic tissue** called the **endoderm**, in relation to a **mesodermal** tissue that protects it. It is essential to understand that the study of the endoderm cannot be done without considering its envelope. Thus, we will examine both the **peritoneum** and the **digestive tract**.

It is important to note that brain development exhibits synchronous phenomena. We will begin by studying the **face**. The embryo undergoes **flexion**, leading to a global reorganisation. Initially, it is an open tissue that delimits itself and organises into specific spaces. We will explore these spaces one by one.

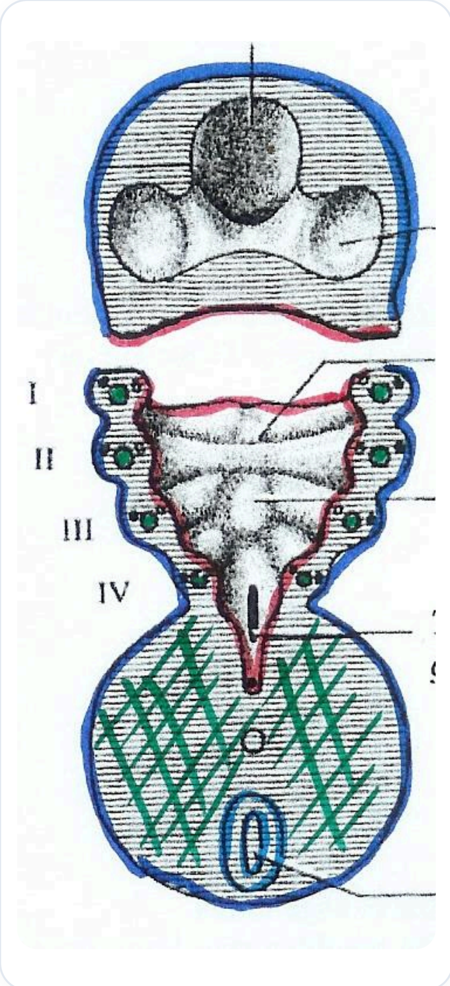

FIG. — Early embryonic development

In our last session, we studied the **neurocranium**. This time, we will focus on the face in detail, from the peripheral part to the internal part, passing through the **mouth**. We will also integrate elements such as the **teeth**, the **pituitary gland**, the **thyroid gland**, the **parathyroid glands**, and the **thymus**, as well as the entire facial plane. The latter organises around **mesodermal** levels such as **Meckel's cartilage** and **Richer's cartilage**.

We will examine the pathway of the **digestive system**, from the mouth, tongue, thyroid, to the lungs, stomach, oesophagus, liver, etc. It is crucial to understand what we digest in our body and how we transform what we eat into **energy**. This process involves fragmentation down to the amino acid level, allowing for reuse by the cellular and molecular system. Waste elimination is also an essential phase of this process.

We have three major sources of elimination phases: through **salts**, through **urine**, and through **respiration**. Indeed, 70% of this elimination occurs via the lungs, which are also an endodermal tissue. To support a lung, it is necessary to work on the colon, as the two are interconnected.

Osteopathy teaches us the importance of anatomy in motion. There is a perpetual movement in the body that never stops. It is essential to accept time, both for tissues and fluids, taking into account the unity between **spirit**, **body**, and **mind**. We will see that the digestive tract is linked to our emotional level and that molecular reactions can influence our way of thinking.

The endoderm is always in relation to the mesoderm, which means that the **visceral system** can influence the cortical system, the parietal system, or the musculoskeletal system. We will examine how this impacts the peritoneum, the sympathetic system, the parasympathetic system, and our nutritional system.

It is crucial to pay attention to **breathing** and **nutrition**. Certain molecules, such as some **prostaglandins**, play a key role in these interactions. The link between the digestive system and the nervous system is probably mediated by the **microbial flora**. We have ten times more bacteria in our gut than cells in our body. We will study this intestinal flora as a system of **fermentation** and **putrefaction**, seeking to maintain a balance in our **allostatic** system.

Fermentation also depends on the seasons. There are four seasons and four inter-seasons, where **fasting** is particularly beneficial. The latter is a period of regeneration for the body. For example, between 15 February and 21 March, we enter a phase of spring ascent.

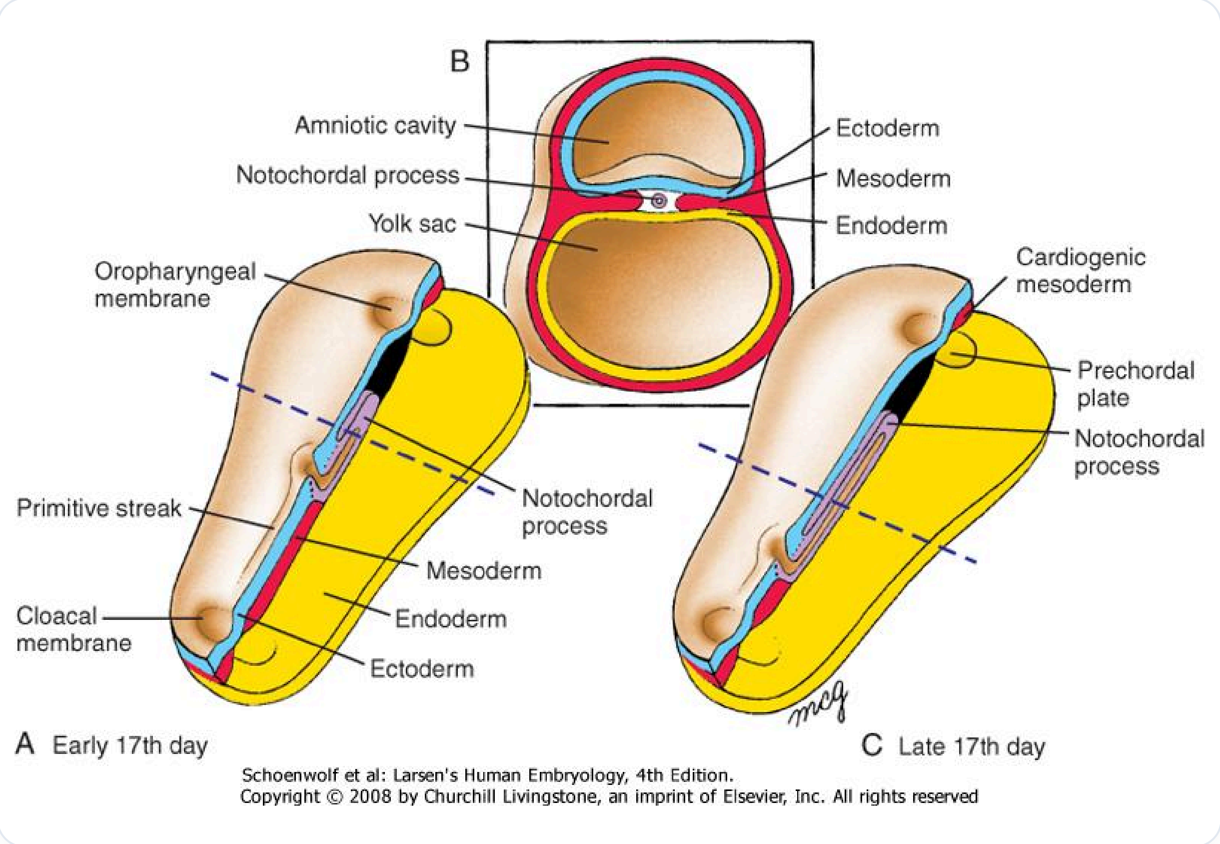

FIG. — Notochordal process

Spring is associated with the colour of the **liver**. Each season has its own characteristics, and the best time to fast is during the inter-seasons. However, it is important to prepare well before fasting, as it releases toxins into the body. Most people lack water, which is essential for interconnection and clarity in our organism.

Hippocrates, during his travels, was concerned about water quality. He tested sources to determine their composition. Observation of the face can also provide information about a person's state of health, in connection with the digestive system and the **neural crest system**.

The face, like the skull, is made of neural crest **ectoderm**, establishing a link between the digestive system and the nervous system. Through precise observation of wrinkles, colours, and shapes, we can obtain indications of a person's vitality.

There are four important therapeutic levels:

1. Bringing people back into their **gravity**.

2. Reconnecting them to their **electromagnetic field**.

3. Bringing them back into their **atomic field**.

4. Restoring them to their **light**.

These elements are essential for regaining the vitality and function for which each person has come. A person's presence can illuminate a room, just as their absence can create a shadow.

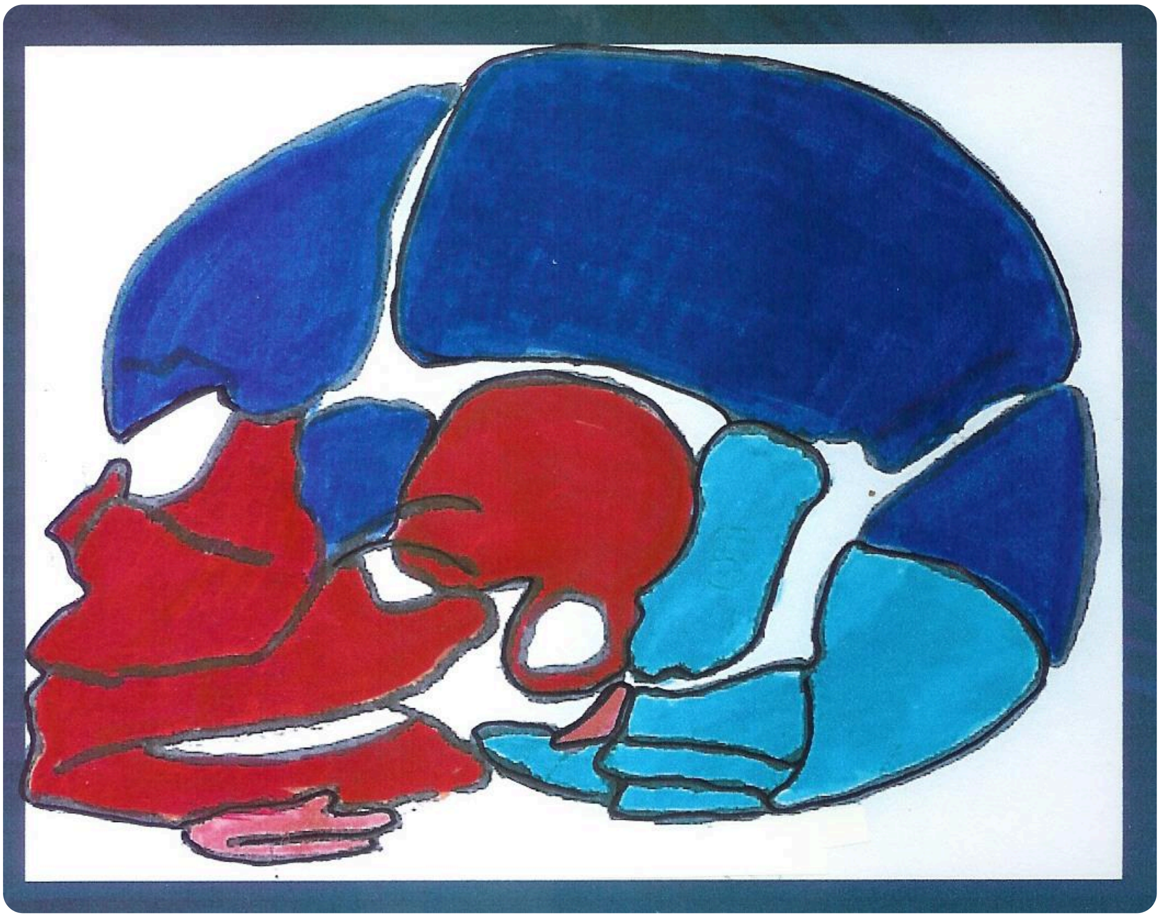

FIG. — Fetal skull

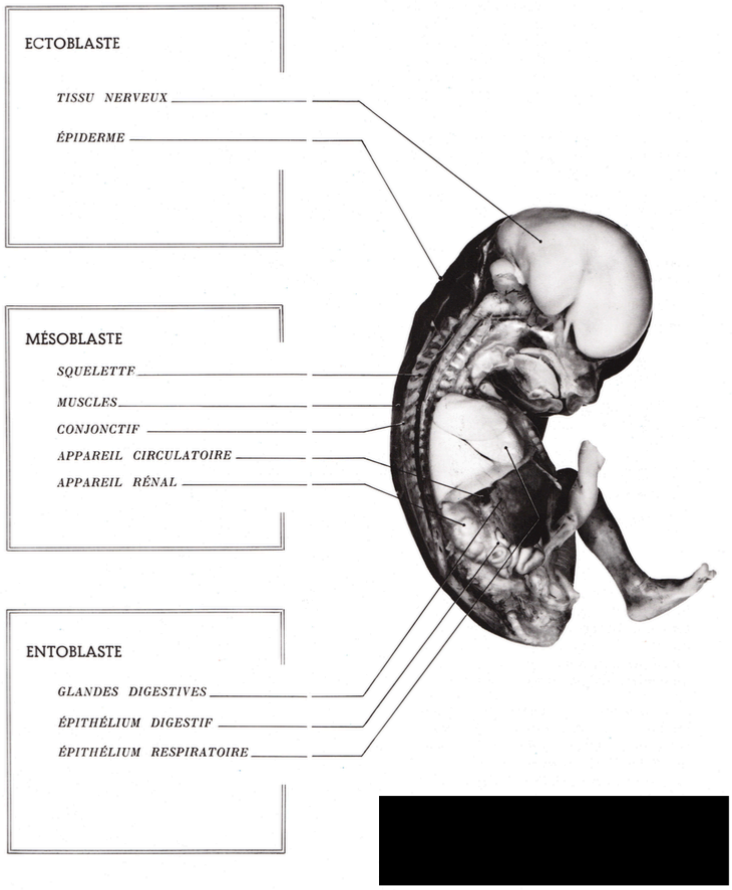

FIG. — *Embryonic germ layers*

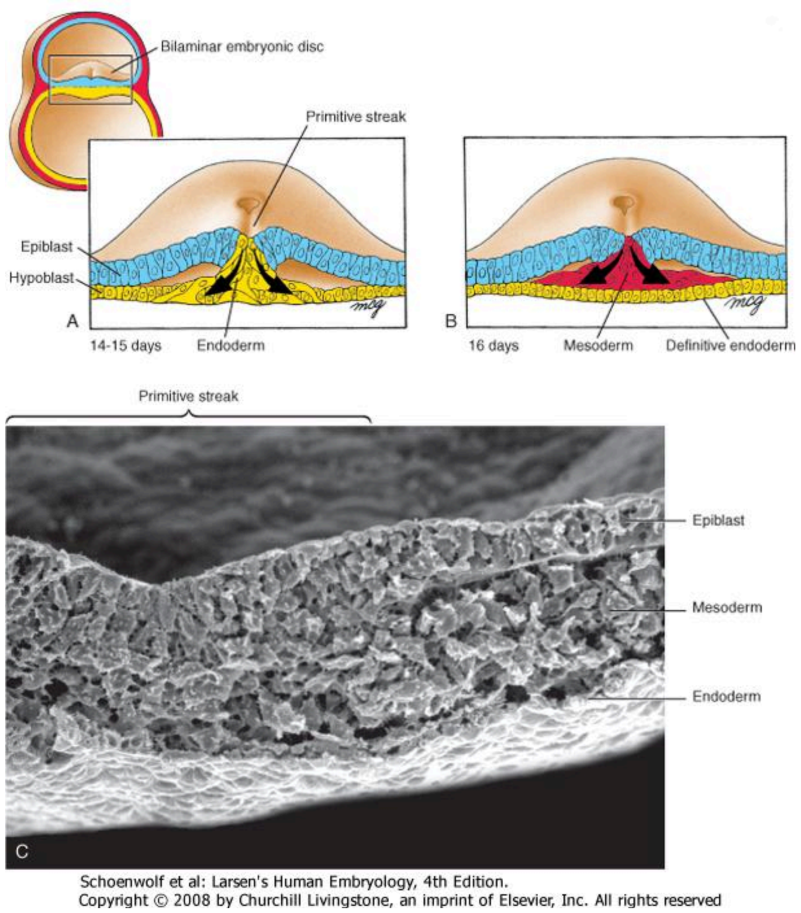

FIG. *— Gastrulation, trilaminar embryonic disc*